Answers/sketch of solutions, to examination in Group and Ring Theory 2021-08-16

1. Set $c = f(1\otimes 1)$. Let $q = \frac{a}{b}$, $r = \frac{d}{e}$, $a, b, d, e \in \mathbb{Z}$, $d, e \neq 0$. Then

$$be\, f(q\otimes r) = be\, f\left(\tfrac{a}{b} \otimes \tfrac{d}{e}\right) = f\left(be \cdot \tfrac{a}{b} \otimes \tfrac{d}{e}\right)$$
$$= f(a\otimes d) = f(ad \cdot 1\otimes 1) = ad\, f(1\otimes 1) = adc,$$

so that $f(q\otimes r) = \dfrac{adc}{be} = cqr$.

2. Let Q be a Sylow-p subgroup of N. Then there exists a Sylow-p subgroup P_1 of G that contains Q. It follows easily that $N \cap P_1 = Q$. An arbitary Sylow-p subgroup P of G is conjugate to P_1, and as N is invariant under conjugation, $N \cap P$ is also a Sylow-p subgroup of N.
   By the second isomorphism theorem $NP/N \simeq P/N\cap P$, and consideration of the orders of G, N, P, $P/N\cap P$ leads to the fact that NP/N is a Sylow-p subgroup of G/N.

3. The invariant factors of A become $x-2$, $(x-2)^2$, and the rational canonical form is hence

$$A \sim \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & -4 \\ 0 & 1 & 4 \end{bmatrix}.$$

4. In $G = S_7$, Sylow-3 subgroups have order $3^2 = 9$. Take
$$P = \langle (123), (456) \rangle$$
$$R = \langle (123), (457) \rangle$$
$$Q = \langle (135), (246) \rangle$$
Then $P \cap Q = \{e\}$ and $P \cap R = \langle (123) \rangle \neq \{e\}$.

5. The sequence of submodules $N_d := (\underbrace{f\circ\cdots\circ f}_{d})(M)$ form a descending chain. As M is artinian there exists n such that $N_n = N_{n+1}$. Let $x \in M$ be arbitrary. There exists then $y \in M$ such that

$$f^n(x) = f^{n+1}(y).$$

As f is injective, the n f's can be "cancelled" one by one, and one obtains $x = f(y)$.

6. By the assumption $\dim_k R < \infty$, R is an artinian ring. As R has no nonzero nilpotent ideals, the Artin-Wedderburn theorem applies, and the commutativity of R leads to

$$R \simeq \prod_{i=1}^{n} k_i$$

with the k_i fields. Then R is the direct sum of the submodules (ideals)

$$I_i = \{0\} \times \cdots \times k_i \times \cdots \times \{0\}, \quad i = 1, \ldots, n,$$

which are all simple R-modules.